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BMJ Open

Osteonecrosis in systemic lupus erythematosus patients and antiphospholipid antibody: a systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046163
Article Type:	Protocol
Date Submitted by the Author:	25-Oct-2020
Complete List of Authors:	Wei, Qijiao; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Zhou, Meng; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Liu, Jianwen; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Chen, Zhihan; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology
Keywords:	RHEUMATOLOGY, IMMUNOLOGY, Musculoskeletal disorders < ORTHOPAEDIC & TRAUMA SURGERY

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Osteonecrosis in systemic lupus erythematosus patients and antiphospholipid antibody: a systematic review protocol

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Word count: 1545.

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ABSTRACT

Introduction: Osteonecrosis (ON) is the destruction of the normal blood supply of bone tissue. ON is the main cause of disability in systemic lupus erythematosus (SLE). Studies have shown that there are many risk factors for SLE complicated by ON including high-dose glucocorticoids, high disease activity and so on. There has been controversy about the correlation between antiphospholipid antibodies (aPLs) and ON in SLE. This study conducted a systematic review of the literature related to SLE aseptic ON and aPLs, and provided a reference for clinical screening of high-risk patients and their early prevention.

Methods and analysis: The following six databases will be searched: MEDLINE/Pubmed, EMBASE, Web of science, the Chinese Biomedical Literature Database (CBM), the Wan-Fang Database, the China National Knowledge Infrastructure (CNKI). A database search will not be restricted by date Case-control study, cohort study or observational studies that compared aLPs between SLE patients with and without ON would be included. Papers in English, Chinese will be included. Two researchers will independently undertake study selection, extraction of data and assessment of study quality. The Newcastle-Ottawa Quality Assessment Scale will be used to assess the quality of the retrieved studies. Meta-analysis will be conducted after screening of studies. Data will be analyzed using odd ratio for dichotomous data.

Ethics and dissemination: Ethical approval is not required because this systematic review will use the published data. The systematic review will be disseminated electronically through a peer-reviewed publication or conference presentations.

Trial registration number: PROSPERO CRD42020209637.

Keywords Antiphospholipid antibody, Antiphospholipid antibody syndrome, Meta-analysis, Osteonecrosis, Systemic lupus erythematosus

Strengths and Limitations of this study:

- This study will systematically report the relationship between SLE aseptic ON and different aPLs, and provided a reference for clinical screening of high-risk patients and their early prevention.

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- The literature will be carefully assessed for quality using the Newcastle-Ottawa Quality Assessment Scale.
- Different types of study design may cause considerable heterogeneity that could limit generating convincing conclusions.

INTRODUCTION

Osteonecrosis (ON) is the destruction of the normal blood supply of bone tissue for various reasons, leading to collapse of the bone structure, causing joint pain and loss of function. The common affected parts are the femoral head and knee joint bone [1]. ON is the main cause of disability in SLE, which seriously affects the life of patients [2]. High-dose glucocorticoids is a definite risk factor for ON, but not the only one. Studies have shown that there are other risk factors for systemic lupus erythematosus complicated by ON [3, 4]. Antiphospholipid antibodies (aPLs) are including lupus anticoagulant (LA), anticardiolipin (ACL) and antiβ2GP1 antibodies. They are closely related to the formation of thrombus [5]. There has been controversy about the correlation between aPLs and ON in systemic lupus erythematosus. Based on these findings, and the fact that no meta-analysis has been introduced. This study conducted a meta-analysis of the literature related to SLE aseptic ON and aPLs, and provided a reference for clinical screening of high-risk patients and their early prevention.

METHODS AND ANALYSIS

This protocol is registered with the International Prospective Register of Systematic Reviews (PROSPERO). We will conform to the PRISMA guidelines for the development of this systematic review.

Research question

The systematic review will aim to answer the following question: what are the presence of aPLs associated with osteonecrosis in SLE?

Patient and public involvement

No patients will be involved.

Criteria for inclusion of studies in this review:

Type of studies

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Observational, case-control studies and cohort studies will be included. Case reports and series (< 5 patients) will be excluded.

Type of participants

SLE patients without limited to age will be included. The studies related to aPLs of SLE combined with aseptic osteonecrosis published in English or Chinese will be included. Cases have clear diagnostic criteria. Patients with other rheumatology diseases including Takayasu arteritis, rheumatoid arthritis, ankylosing spondylitis will be excluded. Patients with other osteonecrosis including traumatic osteonecrosis, infectious osteonecrosis will be excluded.

Type of outcome assessments

Studies provide clear regulations on the sample size. Each literature must provide the odd ratio (OR) and the 95% confidence interval (95% CI).

Search methods for identification of studies

A search strategy will be used and conducted according to the Cochrane handbook guidelines [6]. The following six databases will be searched: MEDLINE/Pubmed, EMBASE, Web of science, the Chinese Biomedical Literature Database (CBM), the Wan-Fang Database, the China National Knowledge Infrastructure (CNKI). The key words include 'Systemic lupus erythematosus', 'Lupus Erythematosus Disseminatus', 'Libman-Sacks Disease', 'Libman Sacks Disease', 'Osteonecrosis', 'Bone Necrosis', 'Avascular Necrosis of Bone', 'Bone Avascular Necrosis', 'Kienbock Disease', 'Kienboeck's Disease', 'kienboecks disease', 'Aseptic Necrosis of Bone', 'Bone Aseptic Necrosis'. The strategy for searching the PUBMED database is shown in **Table 1**. This search strategy will also be applied to the other electronic databases.

Screening and data collection

The title and abstracts will be screened by two authors. Any disagreement during the selection of studies will be discussed and decided by a third author. Details of the selection process are shown in the flow chart (**Figure 1**). Results were screened by checking titles and abstracts for articles to the coexistence of ON, SLE, and aPLs. Studies that compared the rate of aPLs between ON group and non-ON group were

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included.

Quality Assessment and Data extraction

The quality of the included studies was assessed using the Newcastle-Ottawa quality assessment scale (NOS). Scores between 7 and the maximum score of 9 will be defined as high quality, scores between 4 and 6 will be defined as intermediate quality, and scores between 1 and 3 will be defined as low quality. Two authors will independently extract the data and take the following aspects into consideration: general information (name, country, year of publication, title of study and author's publication details, number of patients, mean age and range), study characteristics. All searched studies will be inputted to EndNote software that can assist reviewers to manage data and pick up duplicate publications. These data will then be entered into Stata 11.0 software for analysis. Any disagreement will be discussed and finally decided on by a third author.

Analysis methods

The raw numerical data were extracted from the studies. Meta-analysis was performed using Stata 11.0 software. Before combining the statistics, tests for heterogeneity will be conducted. If $p>0.1$, the fixed-effect model will be used to combine the data. If $p<0.1$, the random-effect model will be used. The reasons leading to heterogeneity will be analyzed and subgroup analysis will be conducted. Results are expressed as ORs with 95% confidence intervals (95% CIs). Two of the authors independently assessed and adjusted the risk of bias. Harbord's modified test and funnel plots were used to assess the reporting biases.

Because our search will include studies across settings, other factors such as glucocorticoid use or hyperlipidemia may confound the independent effect of aPLs on Osteonecrosis in SLE. To tease out the independent effect, we will distinguish and report between descriptive studies and those that adjust for independent effects.

DISCUSSION

Antiphospholipid antibodies (aPLs) are including LA, ACL and β2GP1. The correlation between ON and aPLs in SLE patients was first proposed by Asherson. From then, a large number of studies have investigated the association between them. While the

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results were inconsistent and inconclusive [7, 8]. Then, we reviewed the published studies and undertook a meta-analysis to derive a more precise results.

Studies have shown that ACL, anti-β2GP1 antibody and LA are highly related to thrombosis [9, 10]. Immune complexes and aPLs in SLE patients are important factors that cause blood coagulation [11]. At present, it is believed that ON is a comprehensive result of metabolic factors and local factors affecting blood supply [12, 13]. Thrombosis is a risk factor for ON [14, 15].

The primary objective of this systematic review will be to evaluate the correlation between ON and aPLs in SLE patients. We will conduct a qualitative and quantitative analysis of the overall data included in each study, and we will hopefully find their relationship to provide a reference for clinical screening of high-risk patients and their early prevention.

In conclusion, this is the first systematic review assessing the aPLs association with ON in patients with SLE.. Findings from this review may ascertain the current issues on the topic of aPLs and ON in SLE, and help experts and clinicians reach consensus and develop guidelines to minimise problems and optimise favourable outcomes for patients.

Acknowledgements The authors acknowledge the assistance provided by Fujian Provincial Hospital Library in conducting the literature search.

Author Contributions WQJ prepared the first draft. CZH designed the systematic review protocol and reviewed and revised the first draft. WQJ designed the search strategy and will perform the search. WQJ and ZM will be included in the study screening to extract data and assess the risk of bias in the included studies. LJW and CZH will dispute disagreements between reviewers. WQJ will analyze and interpret the data.

Funding This work was supported by Natural Science Foundation of Fujian Province, Grant No.2019J01184.

Competing interests None declared.

Patient consent for publication Not required.

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Provenance and peer review Not commissioned; externally peer reviewed.

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Table 1 Search strategy used in the Pubmed database.

No.	Search items
1	Systemic lupus erythematosus
2	Lupus Erythematosus Disseminatus
3	Libman-Sacks Disease
4	Libman Sacks Disease
5	or 1-4
6	Osteonecrosis
7	Bone Necrosis
8	Avascular Necrosis of Bone
9	Bone Avascular Necrosis
10	Kienbock Disease
11	Kienboeck's Disease
12	kienboecks disease
13	Aseptic Necrosis of Bone
14	Bone Aseptic Necrosis
15	or 6-14
16	5 and 15

This search strategy will be modified as required for other electronic databases.

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SLE: Systemic lupus erythematosus

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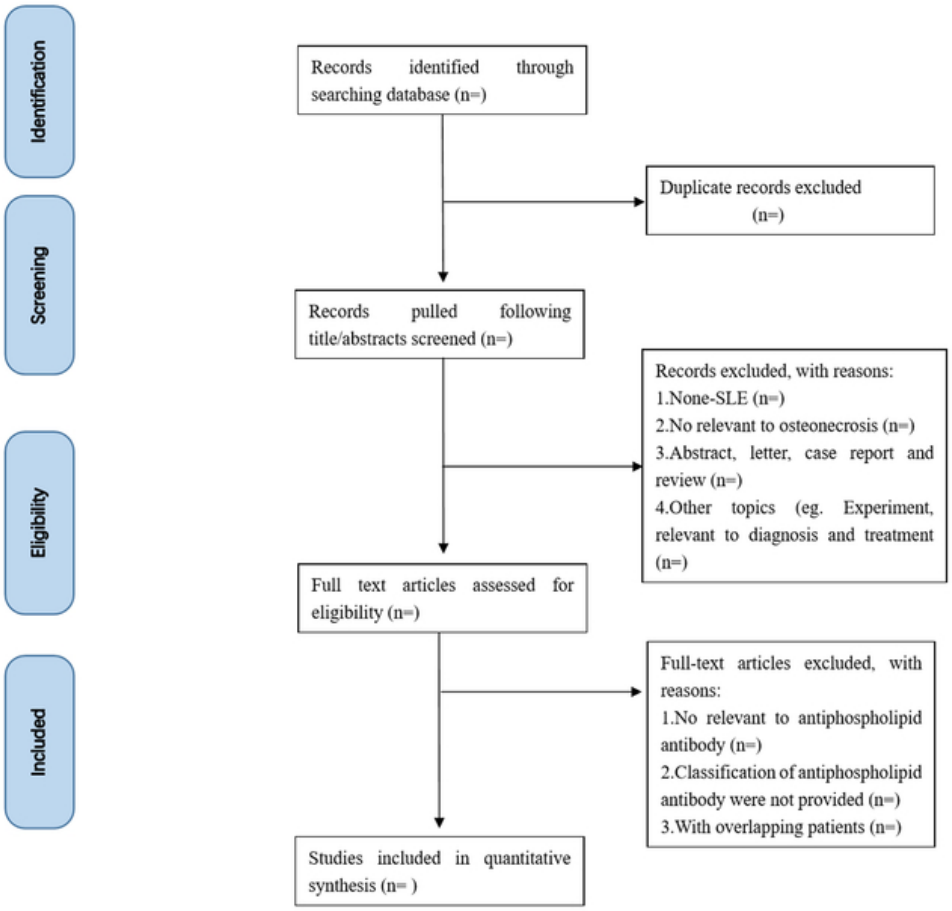


Figure 1 Study flow diagram

28x26mm (600 x 600 DPI)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	2-3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1, 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3-4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	/
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	/
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	/
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	/
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	/
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	/
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	/
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	5
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	6
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	6
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	6

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Relationship between osteonecrosis and antiphospholipid antibodies in patients with systemic lupus erythematosus: A systematic review protocol

Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046163.R1
Article Type:	Protocol
Date Submitted by the Author:	23-Apr-2021
Complete List of Authors:	Wei, Qijiao; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Zhou, Meng; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Liu, Jianwen; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Chen, Zhihan; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology
Primary Subject Heading:	Rheumatology
Secondary Subject Heading:	Immunology (including allergy)
Keywords:	RHEUMATOLOGY, IMMUNOLOGY, Musculoskeletal disorders < ORTHOPAEDIC & TRAUMA SURGERY

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Relationship between osteonecrosis and antiphospholipid antibodies in patients with systemic lupus erythematosus: A systematic review protocol

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Word count (excluding the title page, abstract, references, figures and tables): 1340/4000

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ABSTRACT

Introduction: Osteonecrosis (ON) is characterized by the destruction of the normal blood supply to the bone tissue. ON is the main cause of disability in patients with systemic lupus erythematosus (SLE). Studies have reported the existence of many risk factors for SLE complicated by ON, including the use of high-dose glucocorticoids and high disease activity. The correlation between antiphospholipid antibodies (aPLs) and ON in SLE has been controversial. We aim to conduct a systematic review of the literature related to SLE, aseptic ON, and aPLs, to provide a reference for the clinical screening of high-risk patients and for early prevention.

Methods and analysis: The following six databases will be searched: MEDLINE/PubMed, Embase, Web of Science, Chinese Biomedical Literature Database, Wan-Fang Database, and China National Knowledge Infrastructure. The database searches will not be restricted by date. Case–control studies, cohort studies, or observational studies that compare aPLs between SLE patients with and without ON will be considered eligible. Articles published in English and Chinese will be included. Two researchers will independently perform the processes of study selection, data extraction, and study quality assessment. The Newcastle–Ottawa Quality Assessment Scale will be used to assess the quality of the retrieved studies. A meta-analysis will be performed after screening the studies. Data will be analyzed using odds ratios for dichotomous data.

Ethics and dissemination: Ethical approval is not required because this systematic review will use published data. The systematic review will be electronically disseminated through a peer-reviewed publication or conference presentations.

Trial registration number: PROSPERO CRD42020209637.

Keywords: Antiphospholipid antibody, Antiphospholipid antibody syndrome, Meta-analysis, Osteonecrosis, Systemic lupus erythematosus

ARTICLE SUMMARY:

- This study will systematically report the relationship among systemic lupus

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erythematosus, aseptic osteonecrosis, and different antiphospholipid antibodies, to provide a reference for the clinical screening of high-risk patients and for early prevention.

- The quality of the studies will be carefully assessed using the Newcastle–Ottawa Quality Assessment Scale.
- Different study designs may cause considerable heterogeneity, which could limit the generation of convincing conclusions.

INTRODUCTION

Osteonecrosis (ON) is characterized by the destruction of the normal blood supply to the bone tissue for various reasons, leading to the collapse of the bone structure and causing joint pain and loss of function. The commonly affected parts are the femoral head and knee joints [1]. ON is the main cause of disability in patients with systemic lupus erythematosus (SLE), seriously affecting their quality of life [2]. The use of high-dose glucocorticoids is a definite, but not the only, risk factor for ON. Studies have reported the existence of other risk factors for SLE complicated by ON[3-4]. Antiphospholipid antibodies (aPLs) include lupus anticoagulant (LA), anticardiolipin (ACL), and anti-beta 2 glycoprotein 1 (anti-β2GP1) antibodies, which are closely related to thrombus formation[5]. The correlation between aPLs and ON in SLE has been controversial. On the basis of the previous findings and the fact that no meta-analysis has been conducted, we aim to conduct a meta-analysis of the literature related to SLE, aseptic ON, and aPLs to provide a reference for the clinical screening of high-risk patients and for early prevention.

METHODS AND ANALYSIS

The protocol has been registered with PROSPERO (International Prospective Register of Systematic Reviews). We will conform to the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines for the development of the systematic review.

Research question

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This systematic review aims to answer the following question: What is the relationship between aPLs and ON in SLE?

Patient and public involvement

No patients will be involved.

Criteria for inclusion of studies in this review

Types of studies

Observational, case–control, cohort, and cross-sectional studies will be included. Case reports and cases series (with fewer than five patients) will be excluded.

Types of participants

Patients with SLE will be included without age limitations. Studies related to aPLs in SLE combined with aseptic ON published in English or Chinese will be included. Cases should have clear diagnostic criteria. The diagnosis of SLE should have been made according to the 1982 revised American Rheumatism Association criteria, 1997 American College of Rheumatology classification criteria, Systemic Lupus International Collaborating Clinics classification criteria, or 2019 SLE European League Against Rheumatism/American College of Rheumatology classification criteria. ON should have been radiologically confirmed using plain radiography, bone scans, magnetic resonance imaging, or tomography. Patients with other rheumatologic diseases, including Takayasu arteritis, rheumatoid arthritis, and ankylosing spondylitis, will be excluded. Patients with other types of ON, such as traumatic ON and infectious ON, will be excluded.

Types of outcome assessments

Studies should provide clear regulations on the sample size. Each article must provide the odds ratio (OR) and 95% confidence interval (95% CI) for positivity of any aPL.

Search methods for identification of studies

The search strategy will be applied according to the Cochrane Handbook guidelines[6]. The following six databases will be searched from their inception through December 26, 2020: MEDLINE/PubMed, Embase, Web of Science, Chinese Biomedical Literature Database, Wan-Fang Database, and China National Knowledge

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Infrastructure. The key words will include the following: "Systemic lupus erythematosus," "Lupus Erythematosus Disseminatus," "Libman-Sacks Disease," "Libman Sacks Disease," "Osteonecrosis," "Bone Necrosis," "Avascular Necrosis of Bone," "Bone Avascular Necrosis," "Kienbock Disease," "Kienboeck's Disease," "kienboecks disease," "Aseptic Necrosis of Bone," and "Bone Aseptic Necrosis." The strategy for searching the PubMed database is shown in **Table 1**. This search strategy will also be applied to the other electronic databases.

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The titles and abstracts will be screened by two authors. Any disagreement during the selection of studies will be discussed and decided by a third author. The details of the selection process are shown in the flowchart in **Figure 1**. The results will be screened by checking the titles and abstracts of the articles. Studies pertaining to ON, SLE, and aPLs will be included. Studies that compare the positive rate of aPLs between ON and non-ON groups will be included.

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The quality of the included studies will be assessed using the Newcastle–Ottawa Quality Assessment Scale. Scores between 7 and the maximum of 9 will be defined as high quality; scores between 4 and 6 will be defined as intermediate quality; and scores between 1 and 3 will be defined as low quality. Two authors will independently extract the following data: general information (name, country, year of publication, title of the study, authors' publication details, number of patients, and mean age and range), study characteristics, and follow-up periods (from the measurement of the antibodies to the onset of ON, from the measurement of the antibodies to the end of follow-up, from the onset of ON to the end of follow-up). All searched studies will be imported to EndNote software, which can assist the reviewers in managing data and screening for duplicate publications. The data will subsequently be entered into Stata 11.0 software for analysis. Any disagreement will be discussed and finally decided by a third author.

Analysis methods

Raw numerical data will be extracted from the studies. A meta-analysis will be

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performed, if possible, using Stata 11.0 software. Before combining the statistics, tests for heterogeneity will be conducted. If $p > 0.1$, the fixed-effects model will be used to combine the data. If $p < 0.1$, the DerSimonian and Laird random-effect model will be used. The reasons for heterogeneity will be analyzed, and subgroup analysis will be performed. The subgroup analysis will be performed according to the study design, country of origin of the participants, age of the participants, and sample size, if necessary. The results will be expressed as ORs with 95% CIs. Two of the authors will independently assess the studies and perform adjustments for risk of bias. Harbord's modified test and funnel plots will be used to assess reporting biases.

Because our search will include studies across different settings, other factors such as glucocorticoid use or hyperlipidemia may confound the independent effect of aPLs on ON in SLE. To exclude the independent effect, we will distinguish between descriptive studies and those that adjusted for independent effects.

We hope to complete the entire study by the end of May 2021.

DISCUSSION

aPLs include LA, ACL, and β2GP1. The correlation between ON and aPLs in patients with SLE was first proposed by Asherson. Since then, many studies have investigated this association. However, the results have been inconsistent and inconclusive[7-8]. Therefore, we aim to review the literature and to perform a meta-analysis to derive more precise results.

Previous studies have shown that ACL, anti-β2GP1, and LA antibodies are highly related to thrombosis [9-10]. Immune complexes and aPLs in patients with SLE are important factors that cause blood coagulation[11]. At present, it is believed that ON is a comprehensive result of metabolic and local factors affecting blood supply[12-13]. Meanwhile, thrombosis is a risk factor for ON[14-15].

The primary objective of this systematic review is to evaluate the correlation between ON and aPLs in patients with SLE. We will conduct qualitative and quantitative analyses of the overall data of each study to determine the relationship between aPLs and ON in patients with SLE and, consequently, to provide a reference for the clinical

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screening of high-risk patients and for early prevention.

In conclusion, this will be the first systematic review to assess the association between aPLs and ON in patients with SLE. The findings from this review might clarify the current issues on aPLs and ON in SLE, and may help experts and clinicians reach a consensus and develop guidelines to minimize problems and optimize patient outcomes.

Author Contributions WQJ prepared the first draft. CZH designed the systematic review protocol and reviewed and revised the first draft. WQJ designed the search strategy and will perform the search. WQJ and ZM will be included in the study screening to extract data and assess the risk of bias in the included studies. LJW and CZH will dispute disagreements between reviewers. WQJ will analyze and interpret the data.

Acknowledgements The authors acknowledge the assistance provided by Fujian Provincial Hospital Library in conducting the literature search.

Competing interests None declared.

Funding This work was supported by Natural Science Foundation of Fujian Province, Grant No.2019J01184.

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Table 1 Search strategy used in the PubMed database.

No.	Search items
1	Systemic lupus erythematosus
2	Lupus Erythematosus Disseminatus
3	Libman-Sacks Disease
4	Libman Sacks Disease
5	or 1-4
6	Osteonecrosis
7	Bone Necrosis
8	Avascular Necrosis of Bone
9	Bone Avascular Necrosis
10	Kienbock Disease
11	Kienboeck's Disease
12	kienboecks disease
13	Aseptic Necrosis of Bone
14	Bone Aseptic Necrosis
15	or 6-14
16	5 and 15

This search strategy will be modified as required for other electronic databases.

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Figure legend

SLE：Systemic lupus erythematosus

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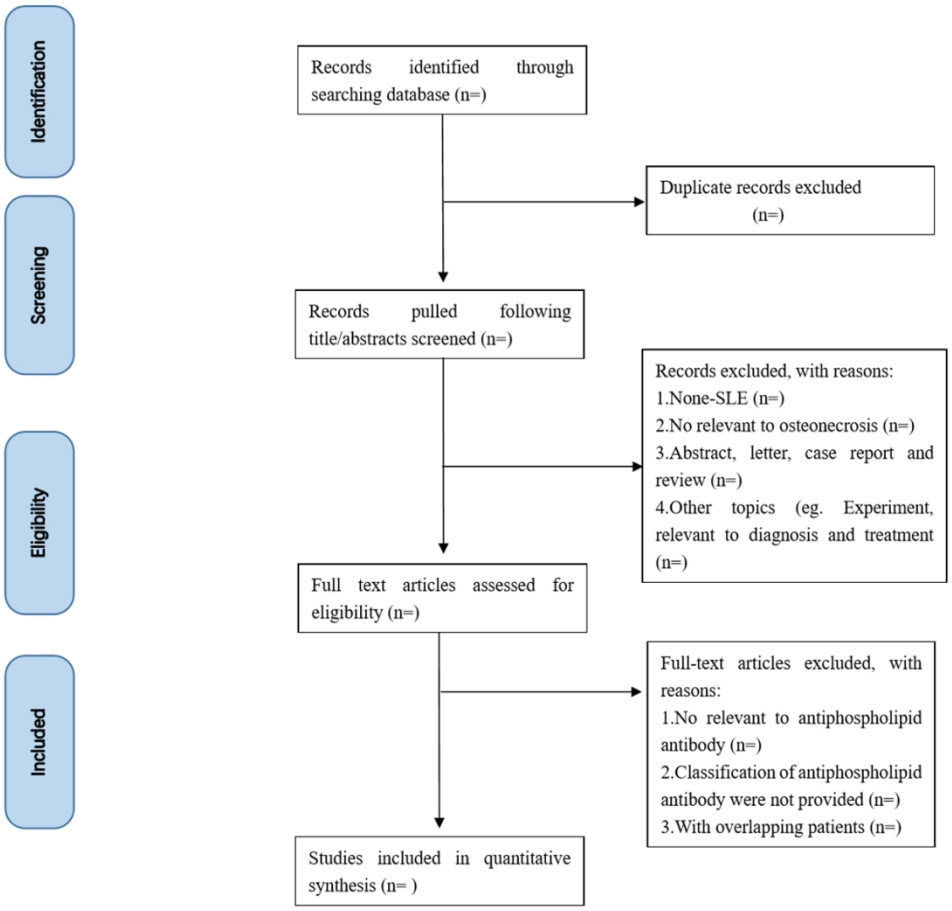


Figure 1 Study flow diagram

28x26mm (1200 x 1200 DPI)

PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	2-3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1, 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3-4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	/
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	/
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	/
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	/
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	/
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	/
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	/
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	5
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	6
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	6
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	6

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(6): e1000097. doi:10.1371/journal.pmed1000097

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Journal:	*BMJ Open*
Manuscript ID	bmjopen-2020-046163.R2
Article Type:	Protocol
Date Submitted by the Author:	12-Jun-2021
Complete List of Authors:	Wei, Qijiao; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Zhou, Meng; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Liu, Jianwen; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology Zhang, Shengli; Fujian Medical University Provincial Clinical Medical College, rheumatology; Fujian Provincial Hospital Gao, Fei; Fujian Medical University Provincial Clinical Medical College, rheumatology; Fujian Provincial Hospital LIn, He; Fujian Medical University Provincial Clinical Medical College, rheumatology; Fujian Provincial Hospital Chen, Zhihan; Fujian Medical University Provincial Clinical Medical College, Rheumatology; Fujian Provincial Hospital, Rheumatology
Primary Subject Heading:	Rheumatology
Secondary Subject Heading:	Immunology (including allergy)
Keywords:	RHEUMATOLOGY, IMMUNOLOGY, Musculoskeletal disorders < ORTHOPAEDIC & TRAUMA SURGERY

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Relationship between osteonecrosis and antiphospholipid antibodies in patients with systemic lupus erythematosus: A systematic review protocol

Wei Qijiao[1,2], Zhou Meng[1,2], Liu Jianwen[1,2], Zhang Sengli[1,2], Gao Fei[1,2], Lin He[1,2], Chen Zhihan[1,2*]

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Word count (excluding the title page, abstract, references, figures and tables): 1340/4000

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ABSTRACT

Introduction: Osteonecrosis (ON) is characterized by the destruction of the normal blood supply to the bone tissue. ON is the main cause of disability in patients with systemic lupus erythematosus (SLE). Studies have reported the existence of many risk factors for SLE complicated by ON, including the use of high-dose glucocorticoids and high disease activity. The correlation between antiphospholipid antibodies (aPLs) and ON in SLE has been controversial. We aim to conduct a systematic review of the literature related to SLE, aseptic ON, and aPLs, to provide a reference for the clinical screening of high-risk patients and for early prevention.

Methods and analysis: The following six databases will be searched: MEDLINE/PubMed, Embase, Web of Science, Chinese Biomedical Literature Database, Wan-Fang Database, and China National Knowledge Infrastructure. The database searches will not be restricted by date. Case–control studies, cohort studies, or observational studies that compare aPLs between SLE patients with and without ON will be considered eligible. Articles published in English and Chinese will be included. Two researchers will independently perform the processes of study selection, data extraction, and study quality assessment. The Newcastle–Ottawa Quality Assessment Scale will be used to assess the quality of the retrieved studies. A meta-analysis will be performed after screening the studies. Data will be analyzed using odds ratios for dichotomous data.

Ethics and dissemination: Ethical approval is not required because this systematic review will use published data. The systematic review will be electronically disseminated through a peer-reviewed publication or conference presentations.

Trial registration number: PROSPERO CRD42020209637.

Keywords: Antiphospholipid antibody, Antiphospholipid antibody syndrome, Meta-analysis, Osteonecrosis, Systemic lupus erythematosus

- **STRENGTHS AND LIMITATIONS:**
- This study will systematically report the relationship among systemic lupus

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erythematosus, aseptic osteonecrosis, and different antiphospholipid antibodies, to provide a reference for the clinical screening of high-risk patients and for early prevention.

- The quality of the studies will be carefully assessed using the Newcastle–Ottawa Quality Assessment Scale.
- Different study designs may cause considerable heterogeneity, which could limit the generation of convincing conclusions.

INTRODUCTION

Osteonecrosis (ON) is characterized by the destruction of the normal blood supply to the bone tissue for various reasons, leading to the collapse of the bone structure and causing joint pain and loss of function. The commonly affected parts are the femoral head and knee joints [1]. ON is the main cause of disability in patients with systemic lupus erythematosus (SLE), seriously affecting their quality of life [2]. The use of high-dose glucocorticoids is a definite, but not the only, risk factor for ON. Studies have reported the existence of other risk factors for SLE complicated by ON[3-4]. Antiphospholipid antibodies (aPLs) include lupus anticoagulant (LA), anticardiolipin (ACL), and anti-beta 2 glycoprotein 1 (anti-β2GP1) antibodies, which are closely related to thrombus formation[5]. The correlation between aPLs and ON in SLE has been controversial. On the basis of the previous findings and the fact that no meta-analysis has been conducted, we aim to conduct a meta-analysis of the literature related to SLE, aseptic ON, and aPLs to provide a reference for the clinical screening of high-risk patients and for early prevention.

METHODS AND ANALYSIS

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Research question

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This systematic review aims to answer the following question: What is the relationship between aPLs and ON in SLE? We only explore the relationship between the positive rate of aPLs and ON, not involving the titer of aPLs.

Patient and public involvement

No patients will be involved.

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Types of studies

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We hope to complete the entire study by the end of May 2021.

DISCUSSION

aPLs include LA, ACL, and β2GP1. The correlation between ON and aPLs in patients with SLE was first proposed by Asherson. Since then, many studies have investigated this association. However, the results have been inconsistent and inconclusive[7-8]. Therefore, we aim to review the literature and to perform a meta-analysis to derive more precise results.

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and ON in patients with SLE and, consequently, to provide a reference for the clinical screening of high-risk patients and for early prevention.

In conclusion, this will be the first systematic review to assess the association between aPLs and ON in patients with SLE. The findings from this review might clarify the current issues on aPLs and ON in SLE, and may help experts and clinicians reach a consensus and develop guidelines to minimize problems and optimize patient outcomes.

Ethics and Dissemination

Ethical approval and patient consent are not required because the results of this this systematic review will be disseminated through publication in a peer-reviewed journal and presented at a relevant conference. The data used in this systematic review will not contain individual patient data. The study commenced at December 26, 2020, and its expected completion date is the end of May 2021. This study aims at reporting the relationship among systemic lupus erythematosus, aseptic osteonecrosis, and different antiphospholipid antibodies, to provide a reference for the clinical screening of high-risk patients and for early prevention.

Author Contributions WQJ prepared the first draft. CZH designed the systematic review protocol and reviewed and revised the first draft. ZM designed the search strategy and will perform the search. LJW and ZM will be included in the study screening to extract data and assess the risk of bias in the included studies. GF, ZSL and CZH will dispute disagreements between reviewers. LH will analyze and interpret the data.

Acknowledgements The authors acknowledge the assistance provided by Fujian Provincial Hospital Library in conducting the literature search.

Competing interests None declared.

Funding This work was supported by Natural Science Foundation of Fujian Province, Grant No.2019J01184.

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Table 1 Search strategy used in the PubMed database.

No.	Search items
1	Systemic lupus erythematosus
2	Lupus Erythematosus Disseminatus
3	Libman-Sacks Disease
4	Libman Sacks Disease
5	or 1-4
6	Osteonecrosis
7	Bone Necrosis
8	Avascular Necrosis of Bone
9	Bone Avascular Necrosis
10	Kienbock Disease
11	Kienboeck's Disease
12	kienboecks disease
13	Aseptic Necrosis of Bone
14	Bone Aseptic Necrosis
15	or 6-14
16	5 and 15

This search strategy will be modified as required for other electronic databases.

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Figure legend

SLE：Systemic lupus erythematosus

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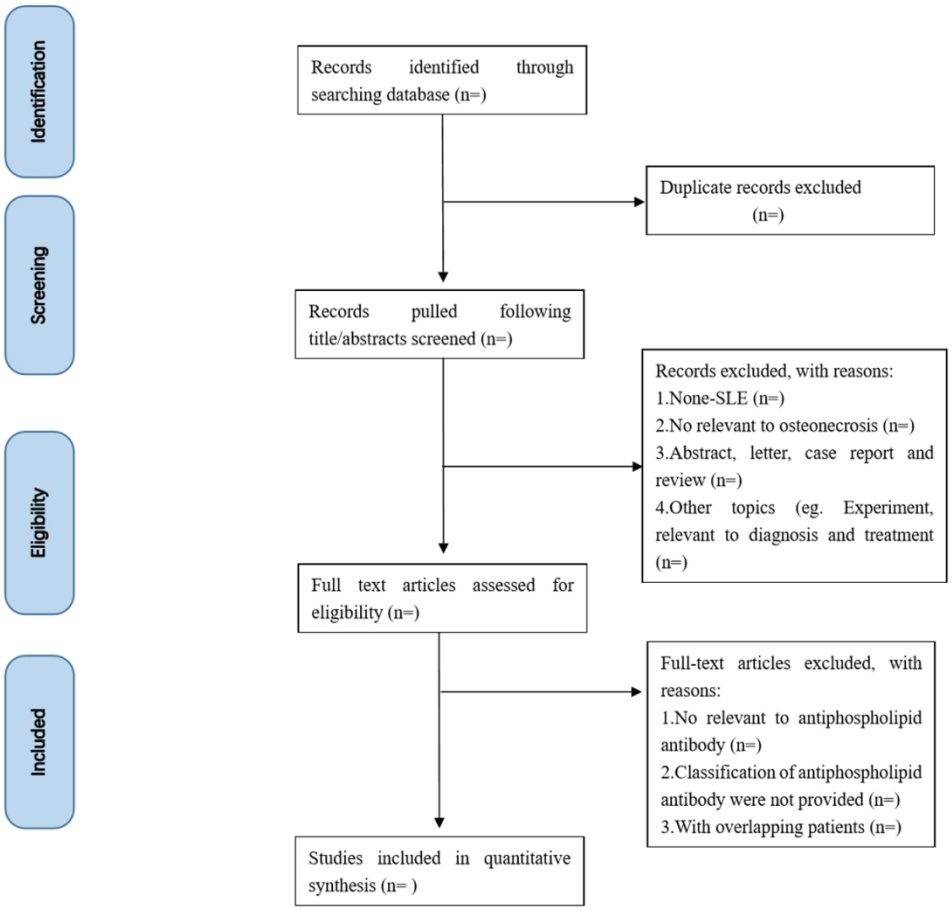


Figure 1 Study flow diagram

28x26mm (1200 x 1200 DPI)



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	2
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	2
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	2-3
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	1, 3
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	3-4
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	4
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	4
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	4-5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	4-5
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	4
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	5
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	5
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	5

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	5
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	5
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	/
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	/
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	/
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	/
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	/
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	/
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	/
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	5
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	6
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	6
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	6

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